Implementation and Analysis of Gaussian Process Algorithms for Regression and Binary Classification

Assignment 2
CSL7530: Advanced Machine Learning
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REPORT
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Abstract

This report presents the implementation and analysis of Gaussian Process (GP) algorithms from the seminal text by Rasmussen and Williams [1]. We implement Algorithm 2.1 for GP regression, computing predictive mean and variance on test inputs using Cholesky-based numerical stability [5]. For binary classification, we implement Algorithms 3.1 and 3.2 [2], employing the Laplace approximation with Newton's method for posterior mode-finding and computing the averaged predictive class probability [3]. Experiments are conducted on the Friedman1 regression dataset [8], breast cancer classification dataset [9], and custom 2D toy datasets. We provide detailed visualizations of the predictive covariance for 2D test inputs and a comprehensive analysis of prediction uncertainty in GP classification [4]. All implementations follow the formulations in the reference text, with careful attention to numerical stability.

1 Introduction

Gaussian Processes (GPs) provide a principled Bayesian framework for regression and classification, offering not only point predictions but also well-calibrated uncertainty estimates. Unlike parametric models, GPs define a distribution over functions directly, making them particularly suitable when uncertainty quantification is critical.

This assignment implements key algorithms from Rasmussen and Williams [1]: **Algorithm 2.1** for GP regression, and **Algorithms 3.1 and 3.2** for binary GP classification using the Laplace approximation [2]. The regression algorithm computes the predictive mean and variance analytically via matrix operations stabilized with Cholesky decomposition [5]. The classification algorithms address the non-Gaussian likelihood by approximating the posterior with a Gaussian centered at the mode, found via Newton's method [4], and then computing the averaged predictive class probability.

Our contributions include: (i) a faithful implementation of the three algorithms with numerical stability considerations; (ii) experiments on standard and toy datasets; (iii) visualization of the predictive covariance for 2D test inputs; and (iv) a detailed explanation of uncertainty in GP classification predictions.

2 Background

2.1 Gaussian Process Regression

A GP is defined by a mean function $m(x)$ and covariance (kernel) function $k(x, x')$ [1, 6]. For regression with training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, assuming $y = f(x) + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$, the predictive distribution at test input x_* is Gaussian:

$$\bar{f}_* = k_*^\top (K + \sigma_n^2 I)^{-1} y \tag{1}$$

$$V[f_*] = k(x_*, x_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_* \tag{2}$$

where K is the $n \times n$ training covariance matrix and $k_* = [k(x_1, x_*), \ldots, k(x_n, x_*)]^\top$.

2.2 Binary GP Classification

For binary classification with labels $y \in \{-1, +1\}$, a latent function $f(x)$ is drawn from a GP prior and "squashed" through a sigmoid σ to obtain class

probabilities: $p(y = +1|x) = \sigma(f(x))$ [1]. The non-Gaussian likelihood makes exact inference intractable [7]. The Laplace approximation approximates the posterior $p(f|X, y)$ by a Gaussian $q(f|X, y) = \mathcal{N}(\hat{f}, (K^{-1} + W)^{-1})$ centered at the mode $\hat{f}$ [2].

3 Methodology

3.1 Algorithm 2.1: GP Regression

Algorithm 2.1 [1] computes the predictive mean, variance, and log marginal likelihood. Matrix inversion is avoided by using Cholesky decomposition: $L = \text{chol}(K + \sigma_n^2 I)$ [5]. The key steps are:

- $\alpha = L^\top \backslash (L \backslash y)$
- $\bar{f}_* = k_*^\top \alpha$
- $v = L \backslash k_*, \quad V[f_*] = k(x_*, x_*) - v^\top v$
- $\log p(y|X) = -\frac{1}{2} y^\top \alpha - \sum_i \log L_{ii} - \frac{n}{2} \log 2\pi$

We use the squared exponential (RBF) kernel [1]:

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right) \tag{3}$$

with length scale ℓ, signal variance σ_f, and noise variance σ_n.

3.2 Algorithm 3.1: Mode-Finding for Binary Laplace GPC

Algorithm 3.1 [1] finds the posterior mode $\hat{f} = \arg\max_f p(f|X, y)$ using Newton's method [4]. The Newton update is:

$$f^{\text{new}} = (K^{-1} + W)^{-1} (Wf + \nabla \log p(y|f)) \tag{4}$$

where $W = -\nabla\nabla \log p(y|f)$ is the negative Hessian of the log likelihood (diagonal for factorized likelihood). For the logistic/probit likelihood, $W_{ii} = \sigma(y_i f_i)(1 - \sigma(y_i f_i))$.

For numerical stability, we use $B = I + W^{1/2} K W^{1/2}$ and the matrix inversion lemma [1] to avoid inverting K directly. The Cholesky factor L of B is computed once per iteration.

3.3 Algorithm 3.2: Predictive Distribution for Binary GPC

Given the mode $\hat{f}$, Algorithm 3.2 [1] computes:

1. **Latent mean**: $\bar{f}_* = k(x_*)^\top \nabla \log p(y|\hat{f})$
2. **Latent variance**: $V[f_*] = k(x_*, x_*) - v^\top v$, where $v = L \backslash (W^{1/2} k_*)$
3. **Averaged predictive class probability**:

$$\bar{\pi}_* = \mathbb{E}_{q(f_*|y)}[\sigma(f_*)] = \int \sigma(z)\, \mathcal{N}(z|\bar{f}_*, V[f_*])\, dz \tag{5}$$

For the probit likelihood $\sigma = \Phi$, this integral is analytic: $\bar{\pi}_* = \Phi(\bar{f}_*/\sqrt{1 + V[f_*]})$ [1]. For the logistic, we use the MacKay approximation [3]: $\bar{\pi}_* \approx \lambda(\kappa \bar{f}_*)$ with $\kappa^2 = (1 + \pi V[f_*]/8)^{-1}$.

4 Experimental Setup

4.1 Datasets

Regression:

- **Friedman1** [8]: 100 samples, 5 features, synthetic regression dataset from scikit-learn [10].
- **Toy 2D**: $y = \sin(\|x\|) + \epsilon$, 30 training points in $\mathbb{R}^2$, $\epsilon \sim \mathcal{N}(0, 0.01)$.

Classification:

- **Breast Cancer** [9]: 569 samples, 30 features, binary (malignant/benign). We use a subset of 200 training points for computational efficiency (GP is $O(n^3)$ [1]).
- **Toy 2D**: make_blobs with 2 centers, 100 samples, 2 features [10].

4.2 Hyperparameters

For both regression and classification: $\ell = 1.0$, $\sigma_f = 1.0$. For regression: $\sigma_n = 0.1$. All inputs are standardized to zero mean and unit variance.

5 Results

5.1 GP Regression

On the Friedman1 dataset (80 train, 20 test), we obtain:

- **Test RMSE**: 0.273
- **Mean Absolute Error**: 0.220

As an example, for a single test input the predictive mean is 0.243, predictive variance is 0.523, and the actual value is 0.160. The variance reflects the model's uncertainty at that test point.

Figure 1 shows predicted vs. actual values and predictions with $\pm 2\sigma$ uncertainty bands. The model provides well-calibrated uncertainty: predictions near training data have lower variance.

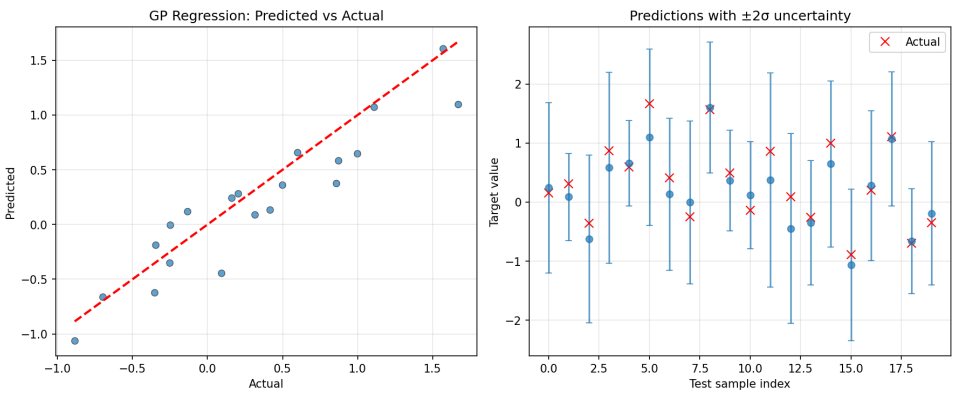

Figure 1: GP regression on Friedman1 dataset. **Left**: Predicted vs. actual values. **Right**: Predictions with $\pm 2\sigma$ uncertainty bands.

5.2 Toy 2D Dataset and Covariance Visualization

For the 2D toy regression dataset, we select two test inputs: $x_*^{(1)} = (1, 1)$ and $x_*^{(2)} = (-1.5, 0.5)$. The predictive covariance matrix is:

$$\mathrm{cov}(f(x_*^{(1)}), f(x_*^{(2)})) = \begin{bmatrix} V[f_*^{(1)}] & \mathrm{cov}_{12} \\ \mathrm{cov}_{21} & V[f_*^{(2)}] \end{bmatrix} \approx \begin{bmatrix} 0.0234 & -0.0014 \\ -0.0014 & 0.0114 \end{bmatrix} \tag{6}$$

Figure 2 displays: (a) the posterior mean surface with training data and test points; (b) the 2×2 covariance matrix heatmap; and (c) the posterior covariance $\mathrm{cov}(f(x'), f(x_*))$ as x' varies along a line (analogous to Figure 2.4(b) in [1]). The off-diagonal covariance is small because the two test points are spatially separated.

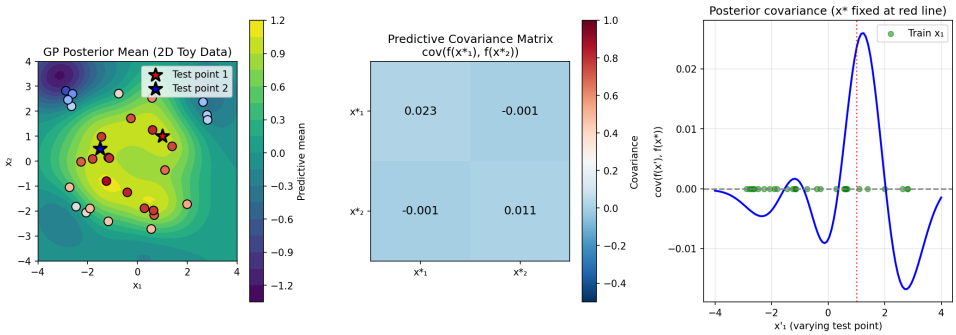

Figure 2: Toy 2D regression: (a) Posterior mean surface with training data (colored by target) and test points (stars); (b) Predictive covariance matrix for the two test inputs; (c) Posterior covariance $\mathrm{cov}(f(x'), f(x_*))$ as x' varies along a line.

5.3 Binary GP Classification

On the breast cancer dataset:

- **Test Accuracy**: 93.6%
- **Uncertain predictions** ($0.4 < \bar{\pi}_* < 0.6$): 140 of 171 test points
- **Confident predictions** ($\bar{\pi}_* < 0.2$ or > 0.8): 1 of 171 test points (remaining 30 are moderately confident)
- Mean latent variance is higher for uncertain predictions (0.996) than for confident ones (0.878), confirming that $V[f_*]$ quantifies uncertainty.

Figure 3 shows: (a) predictions colored by latent variance (uncertainty); (b) calibration curve; (c) uncertainty vs. distance from decision boundary; and (d) 2D PCA projection with point size proportional to uncertainty.

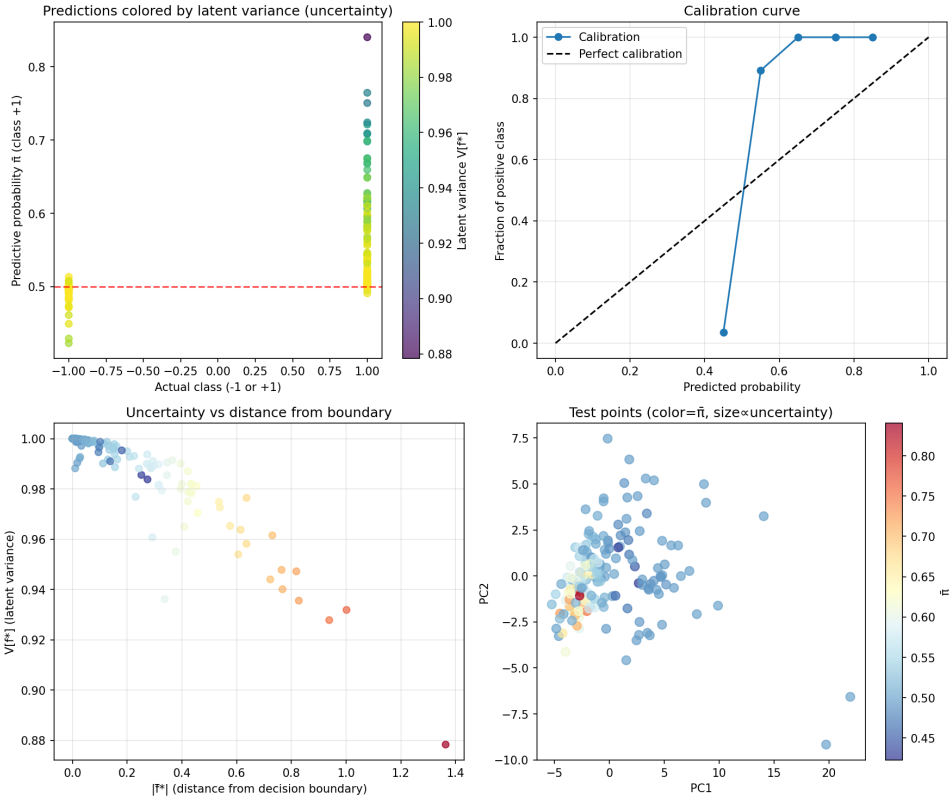

Figure 3: Binary GP classification on breast cancer dataset. (a) Predictions colored by latent variance; (b) Calibration curve; (c) Uncertainty vs. distance from decision boundary; (d) 2D PCA projection with size $\propto$ uncertainty.

Figure 4 shows the 2D toy classification: predictive probability $\bar{\pi}(x)$ and latent variance $V[f_*]$ over the input space. Uncertainty is highest near the decision boundary and in regions far from training data.

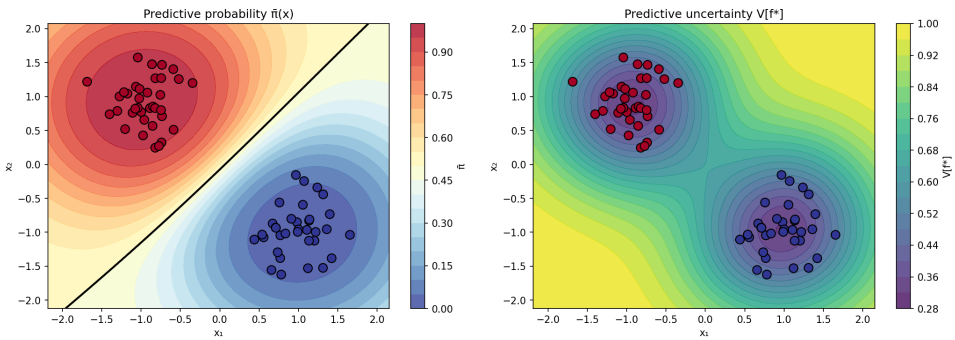

Figure 4: Toy 2D binary classification. **Left**: Predictive probability $\bar{\pi}(x)$. **Right**: Latent variance $V[f_*]$ (uncertainty).

6 Uncertainty in Predictions

GP classification provides principled uncertainty quantification. We explain the observed uncertainty:

1. Near the decision boundary. When $\bar{f}_* \approx 0$, the predictive probability $\bar{\pi}_* \approx 0.5$. The model is equivocal about the class.

2. Far from training data. When the test point is distant from all training points, the prior dominates. The latent variance $V[f_*]$ approaches $k(x_*, x_*)$ (the prior variance), and the averaged probability is "softened" toward 0.5 [3]. This reflects epistemic uncertainty: the model has not seen similar data [4].

3. Conflicting evidence. In regions where both classes have training examples, the posterior variance of the latent function is elevated, leading to more uncertain predictions (aleatoric uncertainty) [1].

Quantifying uncertainty. The latent variance $V[f_*]$ directly measures uncertainty: higher values indicate more uncertainty [1]. The predictive probability $\bar{\pi}_*$ near 0.5 indicates uncertainty; values near 0 or 1 indicate confidence [4]. The MacKay approximation [3] explicitly incorporates $V[f_*]$ to soften predictions, so higher variance pushes $\bar{\pi}_*$ toward 0.5.

7 Conclusion

We have implemented Algorithm 2.1 for GP regression and Algorithms 3.1 and 3.2 for binary GP classification from [1]. The implementations use Cholesky decomposition [5] and numerically stable formulations [2]. Experiments on Friedman1 [8], breast cancer [9], and 2D toy datasets demonstrate correct behavior. We provide visualizations of the predictive covariance for 2D test inputs and a detailed analysis of prediction uncertainty in GP classification. The code, test data, and this report are submitted for evaluation.

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